UNIVERSITÉ PARIS 1 PANTHÉON-SORBONNE

Master 2 — Cybersecurity · Semester 1 · Course S1-ISI5

LECTURE 1

Information Systems Security: Anatomy of an Evolving Battlefield

From foundational architectures to the industrialization of cybercrime and AI-driven risks in 2026.

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Course Overview

This lecture opens the S1-ISI5 module by re-framing information systems (IS) security as a strategic discipline rather than a purely technical one. Its central thesis is that the object we defend — the information system — and the adversaries we defend against have both undergone a structural transformation. The IS is no longer a back-office support function but the operational core of organizational value; the adversary is no longer an isolated individual but an industrialized ecosystem now amplified by artificial intelligence.

The lecture proceeds in three movements. First, it establishes the foundations: what an information system actually is, its six constituent components, and the criteria against which its security is evaluated. Second, it maps the contemporary threat landscape — from the taxonomy of adversaries to the concrete attack vectors that dominate the 2026 environment: Ransomware-as-a-Service, supply-chain compromise, the insider threat, and information warfare. Third, it examines the discontinuity introduced by artificial intelligence and synthesizes the consequences into four governing axioms for the security professional of tomorrow.

HOW TO READ THESE NOTES

Each section corresponds to one analytical building block. Definitions are given in academic form; empirical data points are attributed to their source and should be treated as illustrative orders of magnitude rather than precise constants.

Cross-references to standard frameworks — the CIA triad, ISO/IEC 27001, MITRE ATT&CK, NIS2, GDPR and the EBIOS Risk Manager method — are provided throughout to connect this lecture to the wider curriculum.

Learning outcomes

On completion of this lecture and its associated reading, students should be able to:

1. Argue why information systems security is a determinant of organizational survival and competitiveness, not a discretionary cost centre.
2. Decompose an information system into its six components and explain why security must be holistic across all of them.
3. Apply the confidentiality–integrity–availability criteria (and their extensions) to characterize a security requirement or a breach.
4. Classify threat actors using a motivation × organization matrix and situate real incidents within it.
5. Explain the economic logic of Ransomware-as-a-Service, supply-chain attacks, the insider threat and information-warfare operations.
6. Assess how artificial intelligence operates as a risk multiplier across both offensive deception and offensive scale.
7. Translate the analysis into governance-level recommendations structured around four modern security axioms.

1. The Postulate: The Information System as the Core of Value

The founding postulate of this course is that digital transformation has eradicated the notion of the IS as a mere *"support function."* In a digitalized organization, the information system is the medium through which value is created, exchanged and recorded. Consequently, its security ceases to be a subordinate IT concern and becomes a matter of organizational survival and competitiveness.

This shift has an immediate methodological consequence. If the IS carries the organization's value, then a security failure does not produce a contained technical incident; it produces a business event whose impact propagates simultaneously across operational, financial, reputational and legal dimensions. Four domains of impact make this concrete.

Impacted domain	What is at stake
Operations	Business continuity and the protection of intellectual property. A loss of availability or integrity halts the value chain itself.
Organization	Maintenance of reputation and of the trust of partners. Trust, once lost, is slow and expensive to rebuild.
Employees	Data privacy and a functional working environment. Staff are both a protected population and an operational dependency.
Compliance	Legal mandates and liability — e.g. the NIS/NIS2 directive and the GDPR — which convert security failures into regulatory and financial exposure.

The compliance dimension deserves emphasis for a Master 2 audience. Regulation has turned security from a voluntary good practice into a legal obligation with sanctions. The GDPR governs the protection of personal data across the EU; the NIS2 directive extends binding security and incident-reporting obligations to a broad set of "essential" and "important" entities. *Security governance is therefore inseparable from legal accountability — a theme that recurs in the final axiom of this lecture.*

> **KEY TAKEAWAY**
> If the information system is where value now lives, then protecting it is protecting the organization. Security is a strategic function whose failure is measured in business terms, not in technical terms.

2. Defining the Territory: The Six Components of the IS

Before we can defend the information system we must define its perimeter precisely. An information system is not reducible to its servers or its software; it is a socio-technical ensemble. The lecture identifies six interdependent components, each of which constitutes part of the attack surface.

- **Data —** the most precious asset and therefore the core target of most attacks. Everything else exists to store, move or process it.
- **Applications & Software —** the tools that process data and encode business logic.
- **Networks —** the infrastructure that transports information between components and organizations.
- **Employees —** essential actors, and empirically often the weakest link through which an intrusion begins.
- **Business processes —** the workflows that depend on the IS; their disruption is what turns a technical incident into a business crisis.

- **Facilities —** the physical infrastructure (data centres, premises, power, cabling) housing and sustaining the systems.

The pedagogical point is one of **interdependence**. These components form a densely connected graph: a compromise of one propagates to the others. An attacker who subverts an employee (social engineering) obtains access to applications; through them, to data; and through data, to the business processes that constitute value. Because the components are coupled, security cannot be applied to one in isolation.

> **PRINCIPLE**
> Security must be holistic and coherent, covering the entire ecosystem — data, applications, networks, people, processes and facilities — because the adversary will always target the weakest of the six.

This inventory is also the entry point of structured risk methods. In the **EBIOS Risk Manager** method, for example, the first workshop establishes scope and "business assets" together with the "supporting assets" on which they rely — a formalization of exactly this six-component territory. Naming the components is the precondition for identifying what must be protected and against what.

3. Protecting the Crown Jewels: The Evaluation Criteria

Once the territory is defined, we need criteria against which to evaluate whether it is secure. The classical foundation is the CIA triad — Confidentiality, Integrity, Availability — which expresses the three properties whose preservation constitutes security. Each property is defined by a principle and characterized by the risk incurred when it fails.

Criterion	Principle	Characteristic risk when violated
Confidentiality	Information is accessible only to authorized entities.	Unauthorized disclosure; corporate espionage.
Integrity	Data is accurate, complete and free from unauthorized modification.	Fraud, data manipulation, silent corruption.
Availability	Systems and data are usable on demand by legitimate users.	Business paralysis; Denial of Service (DoS).

For a Master 2 audience it is worth stressing that these three properties are frequently **in tension**. Hardening confidentiality (e.g. aggressive encryption and access lock-outs) can degrade availability; maximizing availability (e.g. broad replication and open access) can erode confidentiality. Security engineering is therefore an exercise in calibrated trade-offs against a defined risk appetite, not in maximizing every property at once.

The triad is also commonly extended. **Traceability / non-repudiation** (the ability to attribute an action to an actor and prevent its later denial) and **authenticity** (proof that an entity or a message is genuinely what it claims to be) are decisive in the 2026 environment, where AI-generated deception directly attacks authenticity — a point developed in Section 10. These criteria are the vocabulary in which security requirements are written and in which breaches are classified throughout the remainder of the course.

> **WHY IT MATTERS**
> The CIA triad gives us a shared, precise language. When we later describe ransomware, a supply-chain compromise or a deepfake, we will characterize each one by which criterion it primarily violates — availability and integrity for ransomware, authenticity and confidentiality for deepfakes and doxing.

4. Mapping the Adversary: The Threat Evolution Matrix

Having defined what we protect and by what criteria, we turn to whom we defend against. A recurring error is to imagine a single, generic "hacker." In reality the adversary population is differentiated and, over three decades, has evolved from isolated amateurs in the 1990s toward the industrialized, structured actors of 2026. A two-axis matrix organizes this population usefully.

- **Vertical axis — Motivation:** from simple curiosity at the low end to financial and destructive intent at the high end.
- **Horizontal axis — Organization:** from isolated / amateur actors on the left to industrial and state-level structures on the right.

Four archetypes populate the resulting quadrants:

Actor	Position	Defining characteristics
Script kiddies	Low motivation, low organization	Limited technical capacity; act mainly to seek recognition. Opportunistic rather than targeted.
Hacktivists & cyber-terrorists	High motivation, low organization	Driven by destabilization, destruction and political messaging rather than profit.
Organized cybercrime	High motivation, high organization	Highly structured operations run under the Ransomware-as-a-Service (RaaS) model. The dominant financially-motivated threat of 2026.
Nation-states	High motivation, high organization	Focused on espionage, sabotage and geopolitical leverage; the best-resourced and most patient adversaries.

The strategic message of the matrix is the rightward and upward drift of the entire population. The centre of gravity of the threat has moved into the high-motivation / high-organization quadrant, and the boundary between organized crime and nation-states is blurring — a hybridization examined in Section 11. Defenders are therefore no longer facing rogue individuals but **structured economies and, increasingly, automated agents**. This reframing is the second of the four axioms in the conclusion.

Methodologically, this matrix corresponds to the "risk origins" (sources de risque) and their "target objectives" analyzed in the second workshop of EBIOS Risk Manager: characterizing the adversary's motivation and resources is a prerequisite for constructing plausible *strategic attack scenarios*.

5. The Industrialization of Extortion: Ransomware-as-a-Service

The single most important development in the criminal threat landscape is the transition from artisanal attacks to an industrialized service economy. The **Ransomware-as-a-Service (RaaS)** model organizes cybercrime along the same division-of-labour logic as a legitimate software business, with specialized roles connected by internal markets.

The RaaS value chain

- **Boss / operators —** manage the core malware and the infrastructure; they own the "product" and the brand.
- **Tool developers —** write the exploits and encryption algorithms.
- **Initial Access Brokers (IABs) —** specialize in breaching networks and selling that access as a commodity.

- **Affiliates —** the intrusion teams that execute attacks using the operators' toolkit, sharing a percentage of the proceeds.

This specialization is what "industrialization" means: no single actor needs to master the entire kill chain. A capability market lowers the barrier to entry, raises the overall volume of attacks, and makes the ecosystem resilient — dismantling one affiliate does not destroy the platform. The economic figures reported in the source deck convey the scale of this economy.

Indicator	Reported figure	Source (as cited)
Ransomware revenue, 2024	**~ $813.55 million**	Chainalysis 2024/2025
Crypto extorted, H1 2025	**~ $460 million**	Chainalysis 2024/2025
Affiliate revenue traced to Russian jurisdictions	**~ 74%**	Chainalysis 2024/2025

Two analytical points follow. First, the geographic concentration of affiliate revenue points to a **geopolitical link**: a large share of proceeds is traced to Russian jurisdictions, which situates "criminal" activity within a tolerant — sometimes instrumentalized — state context, and again blurs the crime/state boundary. Second, students should treat these figures as attributed estimates: ransomware revenue is chronically under-reported (many victims pay silently) and estimates are revised as blockchain analysis matures. The order of magnitude, not the decimal, is the lesson.

WHY IT MATTERS

Because the adversary is a resilient economy rather than a person, defence cannot aim to "catch the hacker." It must raise the cost and lower the yield of the whole business model — hardening initial-access vectors, ensuring recoverable backups (defeating the extortion leverage) and disrupting the payment channel.

6. Permeable Perimeters: The Supply-Chain Vector

Classical security assumed a defensible perimeter separating a trusted interior from a hostile exterior. Digitalization has dissolved that boundary. In 2026 the weakest link is frequently **not inside the organization at all**; it is embedded in the organization's most trusted partnerships. This is the supply-chain attack vector, and its signature is the *domino effect*: a compromise at a distant supplier (Tier 1) cascades through vendors, intermediaries and logistics partners until it reaches the target IS.

Principal sub-vectors

- **Software vendors —** compromising a legitimate software update so that a single poisoned release infects all downstream users simultaneously (the pattern of the SolarWinds-class attacks).
- **Trusted intermediaries —** targeting subcontractors, managed-service providers or telecom operators who legitimately hold access to the target IS.
- **Mergers & acquisitions —** acquiring (or infiltrating) a previously compromised company to obtain a discreet, pre-installed internal entry point.

The mechanism exploits **transitive trust**: organization A trusts supplier B, and B trusts sub-supplier C, so a compromise of C inherits, hop by hop, the access rights that A granted to B. Security spending concentrated on one's own perimeter is therefore structurally insufficient; the effective attack surface is the union of every partner's security posture.

GOVERNANCE RESPONSE

The supply-chain vector is precisely why NIS2 extends obligations to the supply chain and why third-party risk management, contractual security clauses and a "never trust, always verify" (Zero Trust) posture toward partner connections have become mandatory rather than optional. This anticipates the third axiom: trust itself is now an attack surface.

7. The Insider Threat: Access Over Intent

Not every adversary comes from outside. The insider threat is distinctive because it inverts the usual relationship between *intent* and *capability*. An external attacker must acquire access; an insider already possesses it. As the source deck reports, employees are the origin of only around 16% of attacks, yet their potential for critical disruption is disproportionate — and surveys cited indicate that up to 50% of employees would consider corporate sabotage under certain conditions. Low frequency, high severity.

The severity of the insider threat is best understood as the intersection of two forms of privileged knowledge:

- **Technical expertise —** system access and elevated IT privileges (the "keys").
- **Business logic —** knowledge of critical workflows and of exactly where the valuable data resides (the "map").

The **ultimate threat** is the actor who combines both: someone who knows exactly what to break to cause maximum damage and holds the keys to do it. This is why insider risk cannot be managed by perimeter controls; it must be managed through identity governance, least privilege, separation of duties, monitoring of privileged accounts and disciplined off-boarding.

Primary motivations

- **Financial gain —** selling access or data to competitors or to criminal groups (a natural buyer being the Initial Access Brokers of Section 5).
- **Retaliation —** grievance arising from lack of recognition, conflict, or anticipated dismissal.

ANALYTICAL BRIDGE

The insider is the human hinge between the RaaS economy and the supply-chain vector: a bribed or disgruntled employee at a trusted intermediary is simultaneously an insider threat and a supply-chain vulnerability. This is why the modern discipline speaks of managing identities and relationships, not just networks.

8. Information Warfare: E-Reputation and Destabilization

A final category of attack extracts no capital at all. Its objective is to erode the foundation of citizen and market trust — to destabilize rather than to steal. For organizations and states alike, reputation and the perceived integrity of the information environment are themselves assets under attack. The source deck frames this as a **destabilization playbook** with four principal instruments.

Instrument	Mechanism and objective
Distributed Denial of Service (DDoS)	Paralyzing a target's activity to demonstrate its weakness publicly — an attack on availability with a reputational payload.
Data leaks (doxing)	Weaponizing sensitive information for blackmail or political pressure — an attack on confidentiality repurposed for coercion.
Defacement	Altering an organization's public web presence to broadcast ideological messaging — an attack on integrity and image.

Instrument	Mechanism and objective
Synthetic disinformation	Leveraging fake news — increasingly AI-generated — to manipulate diplomacy, elections and corporate valuations. An attack on authenticity at societal scale.

The strategic significance of information warfare is that it targets the **cognitive layer** above the technical one. A DDoS or a defacement may be technically modest yet strategically effective if it undermines confidence in an institution. Synthetic disinformation in particular connects directly to the next part of the lecture: the same generative-AI capabilities that fabricate a convincing news item also fabricate a convincing executive voice. Trust in the authenticity of information is the shared target.

9. The 2026 Catalyst: Artificial Intelligence as a Risk Multiplier

Everything preceding describes an environment that was already industrialized and porous. Artificial intelligence is the catalyst that changes the **shape of the risk curve itself**. The transition is from linear, human-operated intrusion campaigns to exponential, autonomous and highly personalized threat ecosystems — the shift from "manual human attacks" along a flat curve to "autonomous AI attacks" on a steep activation curve.

> **THE CIGREF 2026 INSIGHT**
>
> *"The emergence of cutting-edge AI in cyber-offense is not an incremental evolution… it mathematically and operationally renders traditional defense postures obsolete."*
>
> — as cited in the source deck (CIGREF, 2026)

The claim is strong and deserves careful reading. "Obsolete" does not mean that firewalls or endpoint controls stop functioning; it means that defensive postures **predicated on the tempo and cost of human attackers** lose their assumptions. If reconnaissance, exploitation and social engineering can be produced at machine speed and near-zero marginal cost, then defences calibrated to human-scale attack rates are, quantitatively, outmatched. AI is a risk *multiplier* precisely because it scales existing attack techniques rather than inventing wholly new ones — the vectors of Sections 5–8 are the same, executed faster, cheaper and more convincingly.

The consequence for the defender is a change of regime, not merely of degree: detection and response must themselves move toward automation and machine speed, because a human-in-the-loop response time may be structurally too slow for an autonomous attack. This is the empirical basis for the fourth axiom.

10. AI Attack Vectors: Generative Deception and Excessive Agency

The AI risk profile bifurcates into two complementary paths. The first weaponizes AI's generative capability to attack trust; the second weaponizes AI's autonomy to attack at scale. Together they raise both the quality and the quantity of attacks simultaneously.

Path 1 — Generative deception (targeting trust)

- **Deepfakes & audio cloning —** usurping executive identities to bypass human verification. A cloned CEO voice or video authorizing a transfer defeats controls that ultimately rely on a human recognizing a person — a direct attack on authenticity.

- **Hyper-personalized spear-phishing —** LLM-generated campaigns with flawless grammar and deep contextual knowledge of the target, removing the linguistic and contextual tells that traditionally exposed phishing.

Path 2 — Autonomous agents (targeting scale)

- **Automated reconnaissance —** AI scanning networks to identify vulnerabilities infinitely faster than human teams, compressing the discovery phase of the kill chain.
- **Lateral movement —** AI agents piloting tools, testing credentials and pivoting through environments without requiring a human operator — the phenomenon of "excessive agency," where an autonomous system acts beyond intended boundaries.

The two paths are multiplicative. Generative deception raises the **success probability per attempt** (each lure is more convincing); autonomous agency raises the **number of attempts per unit time and cost**. Their product is the exponential curve of Section 9. Note also that "excessive agency" is a risk not only of the attacker's AI but of the defender's own AI systems: an over-privileged internal agent is itself an attack surface — a theme that connects security to the governance of autonomous algorithms in the final axiom.

> **DEFENSIVE ORIENTATION**
> Countering Path 1 requires strengthening authenticity controls that do not depend on human perception — cryptographic signing, multi-channel verification of high-value instructions, and out-of-band confirmation. Countering Path 2 requires machine-speed detection and strict least-privilege boundaries around every automated identity, human or non-human.

11. Synthesis: The Structural Transformation of Risk

Drawing the threads together, the source deck offers a synthesis attributed to EY (2026): cybersecurity has crossed a **maturity threshold**. It is no longer an isolated technical IT issue but a systemic governance imperative. Systemic cyber risk now sits at the intersection of three forces — geopolitics, digital transformation and economic models — each of which we have seen in operation: geopolitics in the state-tolerated RaaS economy, digital transformation in the dissolution of the perimeter, and economic models in the industrialization of extortion.

The new reality, in three propositions

- **Impacts are transversal —** a single breach simultaneously triggers operational collapse, financial hemorrhage and reputational ruin. There is no longer a purely "technical" incident.
- **Threats are hybridized —** industrialized crime syndicates now wield state-level AI tools, and the crime/state distinction ceases to be operationally meaningful for the defender.
- **The required response is integrated —** resilience demands the combination of executive governance, advanced AI-enabled defensive technologies and strict identity management. No single layer suffices.

The word **systemic** is the operative one. A systemic risk cannot be delegated to a technical team and considered handled; it must be owned at board level, integrated into enterprise risk management, and treated with the same seriousness as financial or geopolitical risk. This is exactly the posture that risk-management methods such as EBIOS Risk Manager institutionalize by linking technical scenarios to business stakes and to a governed treatment plan.

12. Conclusion: Four Axioms of Modern Security Governance

The lecture concludes by compressing its analysis into four axioms. They are deliberately framed as governing principles rather than technical controls, because the central argument of the course is that security has become a governance discipline.

#	Axiom	Implication for the professional
01	**The IS is a living ecosystem**	Protecting isolated technologies is futile; defence must account for data, people, processes and facilities holistically (Sections 2–3).
02	**The adversary is industrialized**	We defend against structured economies and automated agents, not rogue individuals (Sections 4–5, 9–10).
03	**Trust is the primary attack surface**	Vendors, partners and elevated employee identities are the preferred entry vectors. Assume breach (Sections 6–7).
04	**Security is systemic governance**	Tomorrow's professional does not merely secure networks; they govern relationships, permissions, dependencies and autonomous algorithms (Sections 8, 11).

“Assume breach” — the operational corollary of Axiom 3 — is the mindset that unifies the four. It replaces the obsolete goal of a perfect, impermeable perimeter with the realistic goals of **limiting blast radius, detecting quickly and recovering reliably**. A student who leaves this lecture with a single sentence should leave with this one: in 2026 we no longer ask whether the fortress can be breached, but how resiliently the organization operates once it is.

CLOSING STATEMENT

The professional of tomorrow is not only a network defender but a governor of trust — of relationships, permissions, dependencies and, increasingly, of autonomous algorithms. That is the vocation this Master 2 programme prepares you for.

Glossary of Key Terms

Term	Definition
CIA triad	Confidentiality, Integrity, Availability — the three foundational properties whose preservation defines information security.
RaaS	Ransomware-as-a-Service: a criminal business model in which ransomware operators lease their toolkit and infrastructure to affiliates in exchange for a share of proceeds.
Initial Access Broker (IAB)	A specialized criminal actor who breaches networks and sells the resulting access as a commodity to other attackers.
Supply-chain attack	An attack that reaches a target by first compromising a trusted supplier, vendor or intermediary, exploiting transitive trust.
Insider threat	A security risk originating from a person with legitimate access (employee, contractor) who misuses it, deliberately or negligently.
Doxing	The weaponization of leaked sensitive information for blackmail, coercion or reputational damage.
Deepfake	Synthetic audio, image or video generated by AI to convincingly impersonate a real person; an attack on authenticity.
Excessive agency	The risk that an autonomous (AI) system acts beyond its intended boundaries, whether an attacker's agent or an over-privileged internal one.
Zero Trust	A security model that grants no implicit trust based on network location; every access request is authenticated, authorized and continuously verified.
EBIOS Risk Manager	A French (ANSSI) risk-management method that links business stakes, threat sources and attack scenarios to a governed risk-treatment plan.
NIS2	EU directive extending binding cybersecurity and incident-reporting obligations, including supply-chain requirements, to essential and important entities.

Tutorial: Discussion Questions and Assignment

Discussion questions

8. Choose a recent, publicly documented breach. Decompose it across the six components of the IS and identify which the attacker exploited first, and why.
9. Ransomware primarily violates availability and integrity. Argue whether an organization with perfect, tested, offline backups still has a ransomware problem — and what the residual risk is.
10. The deck reports ~74% of affiliate revenue traced to Russian jurisdictions. Discuss the consequences of this geopolitical concentration for law-enforcement and for the crime/state boundary.
11. The insider threat combines technical expertise and business logic. Design three controls that reduce insider risk without degrading a functional working environment (a trade-off from Section 3).
12. Evaluate the CIGREF claim that AI "renders traditional defense postures obsolete." Is this a claim about capability, tempo, or cost? Defend your reading.
13. Reformulate the four axioms as four measurable governance objectives that a CISO could report to a board.

Short written assignment (2–3 pages)

Select one organization (real or fictional but realistic). Using the threat-evolution matrix (Section 4), identify the two adversary archetypes most relevant to it and justify the choice. Then, for one plausible attack scenario, trace the vector (Sections 5–8), state which CIA/authenticity criteria it violates, and propose an integrated response aligned with the four axioms. Where useful, structure the reasoning in the spirit of *EBIOS Risk Manager* (business stakes → risk sources → strategic scenario → treatment).

References and Further Reading

Sources cited in the source lecture deck:

- Chainalysis (2024/2025), Crypto Crime Report — ransomware revenue and extortion figures.
- CIGREF (2026), analysis of AI in cyber-offense (as cited in the source deck).
- EY (2026), synthesis on the structural transformation of cyber risk (as cited in the source deck).

Complementary academic and institutional references:

- ANSSI, EBIOS Risk Manager — the method guide (Agence nationale de la sécurité des systèmes d'information).
- ISO/IEC 27001 and 27005 — Information security management systems and information security risk management.
- MITRE ATT&CK® — knowledge base of adversary tactics and techniques.
- European Union, Directive (EU) 2022/2555 (NIS2); Regulation (EU) 2016/679 (GDPR).
- ENISA, Threat Landscape (annual) — corroborating trend data on ransomware, supply chain and AI-enabled threats.

Lecture notes derived from the S1-ISI5 deck "The Evolving Cyber Battlefield." Empirical figures are reproduced as attributed in the source and should be read as illustrative orders of magnitude.